

PROJECT SPECIFICATION

SECTION 03 30 00 — CAST-IN-PLACE CONCRETE

Al-Noor Hospital Expansion | Atkins Middle East | Spec Ref: AHX-SPEC-0330 Rev 2 | Date: Jan 2024

PART 1 — GENERAL

1.1 SCOPE

This Section covers supply, mixing, delivery, placing, finishing, and curing of cast-in-place concrete for all structural elements including foundations, pile caps, ground beams, suspended slabs, columns, beams, shear walls, and staircase construction. All concrete SHALL comply with BS EN 206:2013+A2:2021 and BS 8500-1:2015+A2:2019 as modified by the project requirements herein.

1.2 SUBMITTALS

[SUBMITTAL REQUIRED] Prior to commencement of any concrete works, the Contractor SHALL submit the following for Engineer's approval (minimum 28 days before use):

- Concrete mix design for each grade proposed, including constituent materials, proportions, water-cement ratio, admixture dosages, and results of trial mixes (minimum 3 No. trial mixes per grade).
- Certificates of conformity for all constituent materials (cement, aggregate, admixtures, water).
- Concrete supplier's quality plan and current third-party certification (BSI, BBA, or equivalent).
- Proposed curing regime and materials for Engineer review.
- Concrete pump and truck mixer calibration records (not older than 12 months).

PART 2 — PRODUCTS

2.1 CONCRETE MIX REQUIREMENTS

Element	Exposure Class (BS EN 206)	Strength Class	Min. Cement (kg/m³)	Max w/c Ratio	Max Cl-	Cement Type
Blinding	XC1	C10/13	240	0.65	0.40%	CEM I or CEM II
Pile caps & gnd beams	XC2/XS1	C30/37	340	0.50	0.20%	SRPC (BS 4027)
Piles	XC3/XS1/XA2	C35/45	380	0.45	0.10%	SRPC (BS 4027)
Suspended slabs	XC3	C35/45	360	0.45	0.40%	CEM I 52.5N
Columns & shear walls	XC3	C35/45	360	0.45	0.40%	CEM I 52.5N
External exposed elements	XC4/XF1	C40/50	380	0.40	0.20%	CEM I 52.5N + Silica Fume

2.2 MATERIALS

CEMENT: Portland cement SHALL conform to BS EN 197-1. Sulphate Resisting Portland Cement (SRPC) where specified SHALL conform to BS 4027. No more than one cement type SHALL be used in any single pour. [MANDATORY]

AGGREGATES: Coarse aggregate SHALL be crushed limestone or granite, maximum size 20mm for general use (10mm for columns and walls). Fine aggregate SHALL be washed natural sand, grading Zone M per BS EN 12620. Sea-dredged aggregate SHALL NOT be used. [MANDATORY]

ADMIXTURES: Only admixtures on the Engineer's approved products list SHALL be used. Calcium chloride admixtures are PROHIBITED in all structural concrete. Combination of admixtures SHALL only be used with supplier's written compatibility certificate. [MANDATORY]

WATER: Potable water complying with BS EN 1008 SHALL be used. Reclaimed or recycled process water SHALL NOT be used without Engineer's written approval. [MANDATORY]

PART 3 — EXECUTION

3.1 PLACING & COMPACTION

Concrete SHALL be placed within 90 minutes of discharge from the mixing plant and before the drum has completed 300 revolutions. [MANDATORY] Concrete SHALL NOT be placed when ambient temperature exceeds 40degC or is below 5degC without written approval and documented hot/cold weather concreting procedures. [MANDATORY] Vibration: Internal poker vibrators SHALL be used at 450mm maximum centres. Vibrator SHALL be inserted vertically and withdrawn slowly at 75mm per second. [MANDATORY]

3.2 TESTING REQUIREMENTS

[TESTING REQUIRED: Slump test (BS EN 12350-2) — minimum 1 per truck, target 160-200mm. Cube test — minimum 4 cubes per 50m³ or per pour, whichever is more frequent. Cubes tested at 7 days (1 cube) and 28 days (2 cubes), 1 cube retained. Characteristic strength compliance: BS EN 206 Clause 8 method. Test facility: UKAS/DAkkS accredited laboratory only.]

3.3 CURING

All concrete SHALL be cured for a minimum period of 7 days. [MANDATORY] Curing SHALL commence immediately after finishing operations are complete. Acceptable methods: (a) Continuous water curing using wet hessian covered by polythene sheeting; (b) Approved curing compound applied at manufacturer's recommended rate. Curing compound SHALL NOT be used on surfaces that are to receive bonded toppings, waterproofing membranes, or epoxy coatings. [MANDATORY]